

STATISTICS MASTER'S COMPREHENSIVE EXAMINATION

2 December 2023

Instructions: Attempt any five out of the following six questions. Specify which five you would like to have graded. Students are allowed calculators, laptops, books and notes. Internet use and cell phones are prohibited. Students are allowed to use laptops for viewing electronic copies of textbooks, but for nothing else.

Problem 1. Let $Y_1, \dots, Y_n$ be i.i.d. samples from the density function

$$f_\theta(y) = \frac{1}{\theta} r y^{r-1} e^{-y^r/\theta} \mathbf{1}_{\theta>0, y>0}.$$

Here $r > 0$ is a known positive constant.

(1) Find the MLE of θ .

(2) Find the limiting distribution of the MLE as $n \rightarrow \infty$.

Answer: The log-likelihood function is given by

$$\begin{aligned} \ell_n(\theta) &\equiv \log L(Y_1, \dots, Y_n | \theta) = \log L(\theta) = \log \left[r^n \theta^{-n} \prod_{i=1}^n Y_i^{r-1} e^{-\theta^{-1} \sum_i Y_i^r} \right] \\ &= -n \log \theta - \frac{1}{\theta} \cdot \sum_i Y_i^r + \text{const.} \end{aligned}$$

Then with $\bar{Y}^r \equiv n^{-1} \sum_i Y_i^r$, the first and second derivatives are

$$\begin{aligned} \ell'_n(\theta) &= -\frac{n}{\theta} + \frac{1}{\theta^2} \sum_i Y_i^r = \frac{n}{\theta} \left(\frac{\bar{Y}^r}{\theta} - 1 \right), \\ \ell''_n(\theta) &= \frac{n}{\theta^2} - \frac{2}{\theta^3} \sum_i Y_i^r = \frac{n}{\theta} \ell'(\theta) - \frac{n \bar{Y}^r}{\theta^3}. \end{aligned}$$

The MLE $\hat{\theta}$ solves that $\ell'(\hat{\theta}) = 0$ so a plausible solution is

$$\hat{\theta} = \bar{Y}^r = \frac{1}{n} \sum_i Y_i^r.$$

This $\hat{\theta}$ is indeed the solution as $\ell''_n(\hat{\theta}) = -n \bar{Y}^r / \hat{\theta}^3 < 0$, i.e. $\hat{\theta}$ is the global maximum of the log likelihood function $\theta \mapsto \ell_n(\theta)$.

Furthermore, the ‘Fisher information’ is given by

$$I(\theta) = -\mathbb{E} \ell''_1(\theta) = -\mathbb{E} \left[\frac{1}{\theta^2} - \frac{2}{\theta^3} Y_1^r \right] = -\theta^{-2} + (2/\theta^3) \mathbb{E} Y_1^r = \theta^{-2}.$$

The last equality follows by computing

$$\begin{aligned} (0.1) \quad \mathbb{E} Y_1^r &= \int y^r f_\theta(y) dy = \int_0^\infty y^r \cdot \frac{1}{\theta} r y^{r-1} e^{-y^r/\theta} dy \\ &= \theta \int_0^\infty (y^r/\theta) e^{-y^r/\theta} d(y^r/\theta) = \theta \cdot \int_0^\infty u e^{-u} du = \theta. \end{aligned}$$

Now by asymptotic theory of the MLE, we have

$$\sqrt{n}(\hat{\theta} - \theta) \rightarrow_d N(0, I^{-1}(\theta)) = N(0, \theta^2).$$

Problem 2. A pharmaceutical manufacturer purchases a particular material from two different suppliers. The mean level of impurities in the raw material is approximately the same for both suppliers, but the manufacturer is concerned about the variability of the impurities from shipment to shipment. If the level of impurities tends to vary excessively for one source of supply, it could affect the quality of the pharmaceutical product. To compare the variation in percentage impurities for the two suppliers, the manufacturer selects ten shipments from each of the two suppliers and measures the percentage of impurities in the raw material for each shipment. The sample means and variances are shown in the accompanying table. Assume the underlying distributions from both suppliers are normal.

Supplier 1	Supplier 2
$\bar{y}_1 = 1.89$	$\bar{y}_2 = 1.85$
$s_1^2 = 0.273$	$s_2^2 = 0.094$
$n_1 = 10$	$n_2 = 10$

- (1) Do the data provide sufficient evidence to indicate a difference in the variability of the shipment impurity levels for the two suppliers? Test using $\alpha = 0.10$. Based on the results of your test, what recommendation would you make to the pharmaceutical manufacturer?
- (2) Find a 90% confidence interval for the population variance for supplier 2, and interpret your results.

Answer: (1). Our test is $H_0 : \sigma_1^2 = \sigma_2^2$ vs. $H_a : \sigma_1^2 \neq \sigma_2^2$, where σ_1^2, σ_2^2 represent the population variances for the two suppliers. The computed test statistic is $F = (.273)/(.094) = 2.904$ with 9 numerator and 9 denominator degrees of freedom. With $\alpha = .05$, $F_{.05} = 3.18$ so H_0 is not rejected: we cannot conclude that the variances are different.

(2). The 90% CI for $\mu_1 - \mu_2$ is (.050, .254). We are 90% confident that the true variance for Supplier B is between .050 and .254.

Problem 3. Investigators apply the weedkiller 2,4-D to 60 lawns; twenty lawns each are treated with 0, 2, and 4 pints per acre. Let x_i represent the amount applied to lawn i , minus 2, so that $\sum_{i=1}^{60} x_i = 0$. Let Y_i represent the the total weight of ground ivy, in pounds, in a square 10 feet on each side in lawn i . Investigators postulate the model $E[Y_i] = \beta_0 + \beta_1 x_i$. The following least-squares parameter estimates are given in Table 1. Furthermore, the residual variance is estimated at 0.84.

- (1) Give an interval designed to contain $E[Y]$ for a lawn treated with 3.2 pints of 2,4-d per acre, with .95 confidence. Make any reasonable assumption about the quantities Y_i , and state the assumption.

Parameter	Estimate	Standard Error
β_0	6.34	0.73
β_1	-2.15	0.84

TABLE 1. Table from problem 3

- (2) Give an interval designed to contain the total ground ivy in the same type of square in a new lawn treated with 3.2 pints of 2,4-d per acre, with .95 confidence.
- (3) Investigators request an interval designed to contain $E[Y]$ for a lawn treated with 6.3 pints of 2,4-d per acre, with .95 confidence. Provide this interval, if the request is sensible, or otherwise give a reason why it is not sensible.

Answer:

- (1) *Assume the measurements have a normal distribution. Then for the lawn in question, estimate $E[Y_i]$ by $6.34 - 2.15 \times 1.2 = 4.534$. Because the x_i values have a zero mean, the estimates of the two regression parameters are independent, and so the standard error of the prediction is $\sqrt{0.73^2 + 0.84^2 \times 1.2^2} = \sqrt{1.549} = 1.245$. Hence an interval for the predicted value is $4.534 \pm 1.96 \times 1.245 = (2.094, 6.974)$.*
- (2) *The prediction for the new lawn is as in the previous part. The standard error for the prediction is $\sqrt{0.84^2 + 1.549} = \sqrt{2.389} = 1.55$. Hence the interval is $4.534 \pm 1.96 \times 1.55 = (1.496, 7.572)$.*
- (3) *The requested interval involves extrapolation well outside the levels of weedkiller represented in the experiment, and, in fact, the prediction is negative, which is impossible. Such a request is not sensible.*

Problem 4. Experimenters investigate the effect of three kinds of fertilizer on three kinds of seeds. They select 45 plots of ground, and on each plot plant an equal number of seeds, with 15 plots containing each of the three seed varieties. They then apply each of the three kinds of fertilizer, to obtain five replicates of each seed type and fertilizer type. They fit a linear model to the yields. The analysis of variance table is Table 2.

	Df	Sum Sq
Seed		16.962
Fertilizer		11.820
Seed:Fertilizer		47.949
Residuals		45.210

TABLE 2. Table from problem 4

- (1) Give the degrees of freedom in the table above.
- (2) Test the null hypothesis that the effect of fertilizer is the same regardless of seed type.

- (3) Test the null hypothesis that fertilizer has no effect, for every type of seed.

Answer:

- (1) 2, 2, 4, 36.
 (2) Use the F test with 4 and 36 degrees of freedom. The statistic is $(47.949/4)/(45.210/36) = 9.545$. The p -value, from the $F_{4,36}$ distribution, is 2.28×10^{-5} .
 (3) Use the F test with 6 and 36 degrees of freedom. The statistic is $((47.949 + 11.820)/(4 + 2))/(45.210/36) = 7.932$. The p -value, from the $F_{6,36}$ distribution, is 1.73×10^{-5} .

Problem 5. A box contains 3 red balls and 2 white balls. We continue to choose balls, at random, without replacement until both white balls have been chosen and then we stop. Let X represent the total number of balls chosen.

- (1) Find $E\{X\}$.
 (2) Find the conditional expected value of X given that the first ball chosen is white.

Answer:

- (1) The variable X takes one of the values 2, 3, 4 or 5. Consider drawing out all 5 balls in order. Then $X = 5$ only if the last draw is white; $P\{X = 5\} = 2/5 = 4/10$. And $X = 4$ only if the last draw is red and the second to last is white; $P\{X = 4\} = (3/5) \times (1/2) = 3/10$. And $X = 3$ only if the last two draws are red and the third to last is white; $P\{X = 3\} = (3/5) \times (1/2) \times (2/3) = 1/5 = 2/10$. And $X = 2$ only if the first two draws are white; this is $(2/5) \times (1/4) = 1/10$. Note that the above probabilities sum to 1, as they should. Then $E\{X\} = (2 + 3 \times 2 + 4 \times 3 + 5 \times 4)/10 = (2 + 6 + 12 + 20)/10 = 4$.
 (2) Conditional on the first ball being white, the variable X takes one of the values 2, 3, 4 or 5. Then $X = 5$ only if the last draw is white; $P\{X = 5|\text{first white}\} = 1/4$. And $X = 4$ only if the last draw is red and the second to last is white; $P\{X = 4|\text{first white}\} = (3/4) \times (1/3) = 1/4$. And $X = 3$ only if the last two draws are red and the third to last is white; $P\{X = 3|\text{first white}\} = (3/4) \times (2/3) \times (1/2) = 1/4$. And $X = 2$ only if the first two draws are white; this is $1/4$. Again the above probabilities sum to 1, as they should. The conditional expected value is $(2 + 3 + 4 + 5)/4 = 7/2$.

Problem 6. We have one continuous random variable X with density function

$$f(x|\theta) = \begin{cases} 2x/\theta^2 & \text{if } 0 < x \leq \theta \\ 0 & \text{Otherwise.} \end{cases}$$

The parameter space of possible θ values is $\Omega = (0, 1)$. Based on X we wish to estimate θ .

- (1) Find $E_\theta\{X\}$ and $Var_\theta\{X\}$.
- (2) Find the maximum likelihood estimator of θ .
- (3) Find an unbiased estimator of $\theta(1 - \theta)$.

Answer:

- (1) $E_\theta\{X\} = \int_0^\theta x 2x/\theta^2 dx = 2x^3\theta^{-2}/3|_0^\theta = 2\theta/3$, and $E_\theta\{X^2\} = \int_0^\theta x^2 2x/\theta^2 dx = 2x^4\theta^{-2}/4|_0^\theta = \theta^2/2$. Hence $Var_\theta\{X\} = E_\theta\{X^2\} - E_\theta\{X\}^2 = \theta^2/2 - 4\theta^2/9 = \theta^2/18$.

- (2) The likelihood is

$$L(\theta) = \begin{cases} 2x\theta^{-2} & \text{if } 0 < x \leq \theta \\ 0 & \text{Otherwise.} \end{cases}$$

Then

$$L'(\theta) = \begin{cases} -4x\theta^{-3} & \text{if } x < \theta < 1 \\ 0 & \text{Otherwise.} \end{cases}$$

Since $L'(\theta) < 0$, L is maximized when θ is at its lowest bound, $\hat{\theta} = x$.

- (3) From earlier parts, $E_\theta\{X\} = 2\theta/3$ and $E_\theta\{X^2\} = \theta^2/2$. Hence $E_\theta\{3X/2 - 2X^2\} = \theta(1 - \theta)$, and so $3X/2 - 2X^2$ is an unbiased estimator of $\theta(1 - \theta)$.